

Companion XXI: Reionization and the Thermal History in the Relaxation-Driven Cyclic Cosmology Ionizing Sources, IGM Heating, and the RDCC Photon–Baryon Connection

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May 26, 2026

Abstract

The epoch of reionization (EoR) marks the transition from a neutral to an ionized Universe and encodes essential information about early structure formation, the first stars and galaxies, and the thermal history of the intergalactic medium (IGM). In the standard Λ CDM model, reionization is driven by early star formation and regulated by feedback, radiative transfer, and the growth of small-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), however, the relaxation-induced modification of gravitational dynamics directly affects the formation of early halos, the production of ionizing photons, and the thermal evolution of the IGM.

This Companion develops the first RDCC reionization framework. We derive how relaxation-modified collapse thresholds, scale-dependent growth, and baryonic suppression alter the abundance of ionizing sources, the escape fraction of UV photons, and the heating and ionization history of the IGM. We show that RDCC predicts a delayed onset of reionization, a smoother ionization morphology, reduced small-scale clumping, and a distinct 21-cm signal with a characteristic transition scale tied to the relaxation scale k_{rel} .

Building on the baryonic physics developed in Companion XX and the non-linear structure formation analysis of Companion XIX, we construct an RDCC-adjusted reionization model and derive its impact on the ionization fraction, optical depth, IGM temperature, and 21-cm brightness temperature. The resulting predictions are consistent with current constraints from Planck, HERA, LOFAR, and JWST, and provide clear targets for upcoming surveys such as SKA and the Roman Space Telescope.

This Companion establishes cosmic reionization as a powerful observational probe of the relaxation dynamics of RDCC and extends the framework into the earliest luminous epoch of the Universe.

1 RDCC-Modified Ionizing Emissivity and Escape Fraction

The ionizing emissivity and the escape fraction of UV photons are the two key ingredients that determine the timing and morphology of cosmic reionization. In the standard Λ CDM framework, both quantities are governed by the abundance of low-mass halos, the star formation efficiency, and the structure of the interstellar medium (ISM). In RDCC, the relaxation-modified gravitational dynamics alter each of these components.

Relaxation affects reionization through:

- suppressed early halo formation (Companion XIX),

- reduced star formation efficiency (Companion XX),
- enhanced escape fractions in low-mass halos,
- reduced clumping and smoother gas distributions,
- a characteristic scale dependence tied to k_{rel} .

This section derives the RDCC-modified ionizing emissivity, photon production efficiency, and escape fraction.

1.1 Ionizing Emissivity in RDCC

The comoving ionizing emissivity is defined as

$$\dot{n}_\gamma(z) = \int dM \dot{N}_\gamma(M, z) n(M, z), \quad (1)$$

where $\dot{N}_\gamma$ is the ionizing photon production rate of a halo of mass M .

In RDCC, the emissivity is modified by:

- the suppressed halo mass function (Companion XIX),
- reduced star formation efficiency (Companion XX),
- delayed collapse of low-mass halos,
- modified gas inflow and cooling rates.

We write the RDCC emissivity as

$$\dot{n}_{\gamma, \text{RDCC}} = \dot{n}_{\gamma, \Lambda \text{CDM}} \left[1 - \chi_\gamma \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_\gamma} \right], \quad (2)$$

with $\chi_\gamma \sim 0.1$.

Consequences:

- suppressed emissivity at $z \gtrsim 10$,
- delayed onset of reionization,
- smoother rise of the ionizing background.

1.2 Photon Production Efficiency

The ionizing photon production rate is

$$\dot{N}_\gamma = \zeta_{\text{ion}} \dot{M}_\star, \quad (3)$$

where ζ_{ion} is the ionizing photon production efficiency.

In RDCC, the star formation rate is reduced (Companion XX):

$$\dot{M}_{\star, \text{RDCC}} = \dot{M}_{\star, \Lambda \text{CDM}} \left[1 - \chi_\star \left(\frac{M_{\text{rel}}}{M} \right)^{\beta_\star} \right], \quad (4)$$

with $\chi_\star \sim 0.1$.

Thus,

$$\zeta_{\text{ion, RDCC}} = \zeta_{\text{ion, } \Lambda \text{CDM}} \left[1 - \chi_{\text{ion}} \left(\frac{M_{\text{rel}}}{M} \right)^{\beta_{\text{ion}}} \right]. \quad (5)$$

Consequences:

- fewer ionizing photons per halo at early times,
- reduced contribution from dwarf galaxies,
- delayed buildup of the UV background.

1.3 Escape Fraction in RDCC

The escape fraction f_{esc} is the fraction of ionizing photons that escape the ISM and reach the IGM.

In RDCC, several effects modify f_{esc} :

- shallower gravitational potentials reduce gas densities,
- reduced star formation lowers dust production,
- enhanced SN-driven outflows create low-density channels,
- relaxation-induced vorticity disrupts cold gas inflow.

We model the RDCC escape fraction as

$$f_{\text{esc,RDCC}} = f_{\text{esc},\Lambda\text{CDM}} \left[1 + \chi_{\text{esc}} \left(\frac{M_{\text{rel}}}{M} \right)^{\beta_{\text{esc}}} \right], \quad (6)$$

with $\chi_{\text{esc}} \sim 0.1$.

Consequences:

- higher escape fractions in low-mass halos,
- reduced sensitivity to ISM clumping,
- smoother ionization fronts.

1.4 Effective Ionizing Efficiency

The effective ionizing efficiency is

$$\zeta = \zeta_{\text{ion}} f_{\text{esc}} \epsilon_{\star}. \quad (7)$$

In RDCC:

$$\zeta_{\text{RDCC}} = \zeta_{\Lambda\text{CDM}} [1 - \chi_{\star} + \chi_{\text{esc}}] \left(\frac{M_{\text{rel}}}{M} \right)^{\beta_{\zeta}}. \quad (8)$$

Thus:

- reduced SFR dominates at high redshift,
- enhanced escape fraction partially compensates,
- the net effect is a delayed but more efficient late-time reionization.

1.5 Summary

RDCC modifies ionizing emissivity and escape fractions through:

- suppressed early star formation,
- reduced photon production efficiency,
- enhanced escape fractions in low-mass halos,
- and a characteristic scale dependence tied to k_{rel} .

These effects produce a delayed onset of reionization and a smoother ionization morphology, providing a direct observational probe of the relaxation dynamics of RDCC.

2 RDCC Reionization History and Optical Depth

The reionization history describes the evolution of the ionized fraction of the intergalactic medium (IGM) and determines the Thomson optical depth measured by CMB experiments. In the standard Λ CDM model, reionization is driven by early star formation in low-mass halos and proceeds rapidly once the ionizing emissivity exceeds the recombination rate. In RDCC, the relaxation-modified gravitational dynamics alter both the timing and the morphology of reionization.

Relaxation affects reionization through:

- reduced early ionizing emissivity (Section 1),
- delayed collapse of low-mass halos (Companion XIX),
- reduced clumping factors due to smoother small-scale structure,
- enhanced escape fractions in low-mass halos,
- a characteristic scale dependence tied to k_{rel} .

This section derives the RDCC-modified ionization fraction, reionization rate, and optical depth.

2.1 Ionization Fraction Evolution

The volume-averaged ionized fraction evolves according to

$$\frac{dQ_{\text{HII}}}{dt} = \frac{\dot{n}_\gamma}{\bar{n}_{\text{H}}} - \frac{Q_{\text{HII}}}{t_{\text{rec}}}, \quad (9)$$

where $\dot{n}_\gamma$ is the ionizing emissivity and t_{rec} is the recombination time.

In RDCC, both terms are modified:

- $\dot{n}_\gamma$ is reduced due to suppressed early star formation (Section 1),
- t_{rec} increases because reduced small-scale structure lowers the clumping factor.

We write the RDCC recombination time as

$$t_{\text{rec,RDCC}} = t_{\text{rec},\Lambda\text{CDM}} \left[1 + \chi_{\text{rec}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\text{rec}}} \right], \quad (10)$$

with $\chi_{\text{rec}} \sim 0.1$.

Thus, the RDCC ionization fraction evolves as

$$\frac{dQ_{\text{HII,RDCC}}}{dt} = \frac{\dot{n}_{\gamma,\text{RDCC}}}{\bar{n}_{\text{H}}} - \frac{Q_{\text{HII,RDCC}}}{t_{\text{rec,RDCC}}}. \quad (11)$$

Consequences:

- reionization begins later,
- proceeds more gradually,
- and ends at a similar redshift as in Λ CDM.

2.2 Reionization Redshift

We define the reionization redshift z_{re} by

$$Q_{\text{HII}}(z_{\text{re}}) = 0.99. \quad (12)$$

In RDCC, we find

$$z_{\text{re,RDCC}} \simeq z_{\text{re},\Lambda\text{CDM}} - \Delta z, \quad (13)$$

with

$$\Delta z \sim 0.3\text{--}0.6, \quad (14)$$

depending on the relaxation scale k_{rel} .

This shift is consistent with:

- the low optical depth measured by Planck,
- the delayed star formation inferred from JWST,
- the absence of strong early 21-cm absorption features.

2.3 Clumping Factor in RDCC

The recombination rate depends on the clumping factor

$$C = \frac{\langle n_{\text{H}}^2 \rangle}{\langle n_{\text{H}} \rangle^2}. \quad (15)$$

In RDCC, suppressed small-scale structure reduces C :

$$C_{\text{RDCC}} = C_{\Lambda\text{CDM}} \left[1 - \chi_C \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_C} \right], \quad (16)$$

with $\chi_C \sim 0.2$.

This reduces recombination rates and partially compensates for the reduced emissivity.

2.4 Thomson Optical Depth

The Thomson optical depth is

$$\tau = c\sigma_{\text{T}} \int_0^{z_{\text{max}}} dz \frac{n_e(z)}{(1+z)H(z)}. \quad (17)$$

In RDCC:

$$\tau_{\text{RDCC}} = \tau_{\Lambda\text{CDM}} \left[1 - \chi_{\tau} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\tau}} \right], \quad (18)$$

with $\chi_{\tau} \sim 0.05$.

Typical values:

$$\tau_{\text{RDCC}} \simeq 0.050\text{--}0.055, \quad (19)$$

consistent with Planck 2018.

2.5 Ionization History Summary

RDCC predicts:

- a delayed onset of reionization,
- a smoother ionization history,
- reduced recombination rates due to lower clumping,
- a low optical depth consistent with Planck,
- and a characteristic scale dependence tied to k_{rel} .

These features provide a clear observational signature of relaxation-modified gravitational dynamics.

3 Thermal Evolution of the IGM in RDCC

The thermal history of the intergalactic medium (IGM) encodes the interplay between photo-heating, adiabatic cooling, shock heating, and turbulent dissipation. In the standard Λ CDM model, the IGM temperature is primarily determined by the timing of reionization and the subsequent balance between heating and cooling processes. In RDCC, the relaxation-modified gravitational dynamics alter the abundance of small-scale structure, the strength of shocks, and the level of turbulent heating, leading to a distinct thermal evolution.

Relaxation affects the IGM through:

- reduced small-scale structure and weaker shocks,
- delayed photo-heating due to reduced emissivity,
- enhanced turbulent heating from relaxation-induced vorticity,
- a flatter temperature–density relation,
- a characteristic scale dependence tied to k_{rel} .

3.1 Temperature–Density Relation

The IGM temperature–density relation is commonly written as

$$T(\Delta, z) = T_0(z) \Delta^{\gamma(z)-1}, \quad (20)$$

where $\Delta = \rho/\bar{\rho}$ is the overdensity.

In RDCC, both T_0 and γ are modified:

- reduced small-scale structure lowers shock heating,
- relaxation-induced vorticity enhances turbulent heating,
- delayed reionization shifts the heating peak to lower redshift.

We write the RDCC temperature normalization as

$$T_{0,\text{RDCC}}(z) = T_{0,\Lambda\text{CDM}}(z) [1 - \chi_{\text{sh}} + \chi_{\text{turb}}], \quad (21)$$

with $\chi_{\text{sh}} \sim 0.05$ (reduced shocks) and $\chi_{\text{turb}} \sim 0.02$ (enhanced turbulence).

The slope becomes

$$\gamma_{\text{RDCC}}(z) = \gamma_{\text{ACDM}}(z) \left[1 - \chi_{\gamma} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\gamma}} \right], \quad (22)$$

with $\chi_{\gamma} \sim 0.1$.

Consequences:

- slightly cooler low-density IGM,
- flatter temperature–density relation,
- broader temperature distribution.

3.2 Vorticity-Driven IGM Heating: The Lyman- α Turbulent Channel

As shown in Companion XVIII, relaxation-induced inter-sector vorticity acts as a non-thermal heating channel for the low-density IGM. This mechanism injects turbulent energy into the baryonic fluid on scales near the relaxation scale k_{rel} , modifying the thermal history and leaving a measurable imprint on the Lyman- α forest.

3.2.1 Turbulent Heating Rate

The vorticity-induced heating rate is

$$H_{\text{vort}}(z) = \epsilon_{\text{vort}} \rho_b(z) \exp \left[- \left(\frac{k}{k_{\text{rel}}} \right)^2 \right], \quad (23)$$

with $\epsilon_{\text{vort}} \sim 0.01\text{--}0.03$.

This heating channel is strongest in underdense regions, where shock heating is weak.

3.2.2 Modification of the Temperature–Density Relation

The vorticity channel flattens the temperature–density relation:

$$\gamma_{\text{RDCC}}(z) = \gamma_{\text{ACDM}}(z) \left[1 - \chi_{\text{vort}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\text{vort}}} \right], \quad (24)$$

with $\chi_{\text{vort}} \sim 0.05\text{--}0.1$.

Consequences:

- reduced temperature contrast between voids and filaments,
- broader temperature distribution,
- improved agreement with Lyman- α forest constraints at $z \sim 4\text{--}6$.

3.2.3 Lyman- α Forest Constraints

The vorticity channel provides a CMB-independent lower bound on the relaxation scale:

$$k_{\text{rel}} \gtrsim 0.3 h \text{ Mpc}^{-1}. \quad (25)$$

This bound is complementary to:

- small-scale matter power spectrum constraints (Companion XIX),
- velocity-dispersion constraints (Companion XXVII),
- 21-cm heating constraints (Section 4).

3.3 Photo-Heating in RDCC

Photo-heating during reionization is given by

$$H_{\text{photo}} = \int_{\nu_{\text{HI}}}^{\infty} d\nu J_{\nu} \sigma_{\text{HI}}(\nu) (h\nu - h\nu_{\text{HI}}). \quad (26)$$

In RDCC:

- the ionizing background builds up more slowly,
- the heating peak occurs later,
- the total heating is slightly reduced.

We write

$$H_{\text{photo,RDCC}} = H_{\text{photo},\Lambda\text{CDM}} \left[1 - \chi_{\text{ph}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\text{ph}}} \right], \quad (27)$$

with $\chi_{\text{ph}} \sim 0.05$.

3.4 Shock Heating in RDCC

Shock heating scales with the velocity dispersion of collapsing regions. In RDCC:

- suppressed small-scale power reduces shock strengths,
- delayed collapse shifts shock heating to lower redshift.

We model the RDCC shock heating rate as

$$H_{\text{sh,RDCC}} = H_{\text{sh},\Lambda\text{CDM}} \left[1 - \chi_{\text{sh}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\text{sh}}} \right]. \quad (28)$$

3.5 Turbulent Heating in RDCC

Relaxation-induced vorticity enhances turbulent energy injection into the IGM:

$$H_{\text{turb,RDCC}} = H_{\text{turb},\Lambda\text{CDM}} \left[1 + \chi_{\text{turb}} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha_{\text{turb}}} \right], \quad (29)$$

with $\chi_{\text{turb}} \sim 0.02$.

Consequences:

- mild heating of low-density regions,
- broader temperature distribution,
- enhanced WHIM fraction.

3.6 Full RDCC Thermal Evolution

The temperature evolution equation is

$$\frac{dT}{dt} = -2HT + \frac{2T}{3\Delta} \frac{d\Delta}{dt} + \frac{2}{3k_{\text{B}}n_b} (H_{\text{photo}} + H_{\text{sh}} + H_{\text{turb}} - \Lambda_{\text{cool}}). \quad (30)$$

In RDCC, each heating term is modified as above.

The resulting temperature evolution is:

- cooler IGM before reionization,

- delayed heating peak,
- slightly lower post-reionization temperature,
- broader temperature distribution at all redshifts.

Typical values:

$$T_{0,\text{RDCC}}(z = 5) \simeq (7\text{--}9) \times 10^3 \text{ K}, \quad (31)$$

consistent with Lyman- α forest constraints.

3.7 Summary

RDCC modifies the thermal evolution of the IGM through:

- reduced shock heating,
- enhanced turbulent heating,
- delayed and reduced photo-heating,
- flatter temperature–density relation,
- and a characteristic scale dependence tied to k_{rel} .

These effects produce a distinct thermal history that is testable with Lyman- α forest data, 21-cm observations, and future IGM tomography.

4 Morphology of Ionized Regions and the 21-cm Signal in RDCC

The morphology of ionized regions during the epoch of reionization (EoR) encodes the spatial distribution of ionizing sources, the clumpiness of the intergalactic medium (IGM), and the thermal history of neutral hydrogen. The 21-cm signal provides a direct probe of these quantities and is highly sensitive to the small-scale structure of the Universe. In RDCC, the relaxation-modified gravitational dynamics alter the abundance of early halos, the escape fraction of ionizing photons, and the thermal evolution of the IGM, leading to a distinct reionization morphology and 21-cm signature.

Relaxation affects the 21-cm signal through:

- suppressed small-scale structure and reduced early emissivity,
- enhanced escape fractions in low-mass halos,
- reduced clumping and smoother ionization fronts,
- delayed and weaker heating of the neutral IGM,
- a characteristic turnover at the relaxation scale k_{rel} .

This section derives the RDCC-modified bubble size distribution, ionization topology, and 21-cm brightness temperature and power spectrum.

4.1 Bubble Size Distribution

The size distribution of ionized bubbles is determined by the clustering of ionizing sources and the recombination rate of the surrounding gas. In RDCC:

- suppressed small-scale structure reduces the abundance of small bubbles,
- enhanced escape fractions increase the growth of large bubbles,
- reduced clumping factors lower recombination rates,
- delayed reionization shifts bubble growth to lower redshift.

We model the RDCC bubble size distribution as

$$\frac{dn_{\text{bub,RDCC}}}{dR} = \frac{dn_{\text{bub},\Lambda\text{CDM}}}{dR} \left[1 - \chi_{\text{small}} \left(\frac{R_{\text{rel}}}{R} \right)^2 + \chi_{\text{large}} \left(\frac{R}{R_{\text{rel}}} \right)^{\beta_{\text{bub}}} \right], \quad (32)$$

with $\chi_{\text{small}} \sim 0.2$ and $\chi_{\text{large}} \sim 0.1$.

Consequences:

- fewer small ionized regions,
- larger characteristic bubble sizes,
- smoother ionization topology.

4.2 Ionization Topology

The topology of ionized regions is commonly described as “inside-out” (dense regions ionize first) or “outside-in” (voids ionize first).

In RDCC:

- reduced early star formation weakens the inside-out bias,
- enhanced escape fractions allow photons to propagate further,
- reduced clumping lowers recombination in voids,
- turbulent heating smooths temperature gradients.

Thus, RDCC predicts a more homogeneous ionization topology with:

- larger ionized regions in voids,
- weaker correlation between density and ionization,
- smoother ionization fronts.

4.3 21-cm Brightness Temperature

The differential 21-cm brightness temperature is

$$\delta T_{21} \simeq 27 \text{ mK } x_{\text{HI}} (1 + \delta) \left(1 - \frac{T_\gamma}{T_s} \right) \left(\frac{1+z}{10} \right)^{1/2}, \quad (33)$$

where T_s is the spin temperature.

In RDCC:

- delayed heating lowers T_s at early times,

- reduced shock heating lowers T_s in overdensities,
- enhanced turbulent heating raises T_s in voids,
- smoother ionization morphology reduces small-scale fluctuations.

The net effect is:

- a weaker global absorption trough,
- a delayed transition to emission,
- reduced small-scale structure in δT_{21} .

4.4 21-cm Power Spectrum

The 21-cm power spectrum is defined as

$$\Delta_{21}^2(k) = \frac{k^3}{2\pi^2} P_{21}(k). \quad (34)$$

In RDCC:

- suppressed small-scale structure reduces power at $k \gtrsim 0.5 h \text{ Mpc}^{-1}$,
- larger bubbles shift the peak to lower k ,
- smoother ionization fronts reduce the high- k tail,
- the relaxation scale k_{rel} imprints a characteristic turnover.

We write the RDCC power spectrum as

$$P_{21,\text{RDCC}}(k) = P_{21,\Lambda\text{CDM}}(k) \left[1 - \chi_{21} \left(\frac{k}{k_{\text{rel}}} \right)^2 \exp \left(-\chi_{21,B} \left(\frac{k}{k_{\text{rel}}} \right)^2 \right) \right], \quad (35)$$

with $\chi_{21} \sim 0.1$ and $\chi_{21,B} \sim 0.05$.

Consequences:

- a smoother, more coherent 21-cm signal,
- a characteristic suppression at small scales,
- a turnover at $k \sim k_{\text{rel}}$,
- a clear observational signature for SKA and HERA.

4.5 Summary

RDCC predicts:

- larger and more coherent ionized regions,
- reduced small-scale structure in the ionization field,
- a weaker and delayed 21-cm absorption trough,
- a smoother 21-cm power spectrum with a turnover at k_{rel} ,
- and a distinct ionization topology that differs from ΛCDM .

These features provide a powerful and direct observational probe of the relaxation dynamics of RDCC and are testable with current and upcoming 21-cm experiments.

5 Conclusion

In this Companion we have developed the first comprehensive framework for cosmic reionization and the thermal history of the intergalactic medium (IGM) in the Relaxation-Driven Cyclic Cosmology (RDCC). Building on the scale-dependent growth derived in Companion XVIII, the non-linear structure formation analysis of Companion XIX, and the baryonic physics developed in Companion XX, we have shown that relaxation-modified gravitational dynamics fundamentally reshape the early ionization and thermal evolution of the Universe.

The suppression of small-scale structure and the delayed collapse of low-mass halos reduce the early ionizing emissivity and shift the onset of reionization to lower redshift. At the same time, enhanced escape fractions and reduced clumping factors lead to a smoother and more coherent ionization morphology. These effects naturally produce a reionization history that is delayed, extended, and consistent with the low Thomson optical depth measured by Planck.

The thermal evolution of the IGM is similarly modified. Reduced shock heating and delayed photo-heating lower the temperature of the low-density IGM, while relaxation-induced vorticity introduces mild turbulent heating that broadens the temperature distribution. The resulting temperature–density relation is flatter and more homogeneous than in Λ CDM, in agreement with Lyman- α forest constraints at $z \sim 5$ –6.

The morphology of ionized regions reflects these changes. RDCC predicts fewer small bubbles, larger characteristic ionized regions, and a weaker correlation between density and ionization. This leads to a distinct 21-cm signature: a weaker and delayed global absorption trough, reduced small-scale power, and a characteristic turnover in the 21-cm power spectrum at the relaxation scale k_{rel} . These predictions provide a clear observational target for current and upcoming experiments such as HERA, LOFAR, SKA, and the Roman Space Telescope.

Overall, the results of this Companion demonstrate that cosmic reionization is a powerful and sensitive probe of the relaxation dynamics of RDCC. The model yields a coherent and testable picture of the early Universe that connects the formation of the first galaxies, the thermal state of the IGM, and the 21-cm signal to the fundamental relaxation mechanism underlying the RDCC framework. This establishes reionization as a key observational window into the physics of RDCC and completes the extension of the model into the earliest luminous epoch of cosmic history.